

## Lab 1: Requirements Engineering & UML Use-Case Modelling

**Student Name:** Saatvik Gupta  
**SRN:** PES1UG24CS395  
**Section / Semester:** G / V Semester  
**GitHub Repository Link:** [https://github.com/Saatvik-G/SE-LABS\\_PES1UG24CS395.git](https://github.com/Saatvik-G/SE-LABS_PES1UG24CS395.git)

### 1. Requirements Specification Table

| ID      | Type           | Description                                                                                                                                                           | Pri    | Acceptance Criteria                                                                                                                                                                   | Rationale                                                                                                |
|---------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| FR-001  | Functional     | The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage.                                     | High   | Pass: Pallet assigned to valid rack;<br>Fail: Placement permitted on rack where total weight exceeds rated structural limit.                                                          | Ensures physical safety of the warehouse by preventing rack collapses due to overload.                   |
| FR-002  | Functional     | The system shall log stock movements automatically when a barcode or RFID tag is scanned by a Warehouse Operator.                                                     | High   | Pass: System records timestamp, SKU, operator ID, and origin/destination coordinates upon successful scan;<br>Fail: Scan does not register or logs incomplete details.                | Enables real-time tracking of inventory movements and maintains stock accuracy.                          |
| FR-003  | Functional     | The system shall allow a Logistics Supervisor to configure and modify the 3D coordinate layout and maximum weight capacity of warehouse racks.                        | Medium | Pass: Updated weight limits are immediately applied to pallet validation logic;<br>Fail: Non-supervisors can modify limits, or updates fail to reflect in validations.                | Allows the supervisor to adapt the storage layout based on structural changes or product specifications. |
| FR-004  | Functional     | The system shall provide a search interface that allows operators to look up the 3D coordinates (zone, aisle, shelf, position) of a pallet using its SKU code.        | High   | Pass: Search returns exact 3D coordinates for active SKUs and returns 'SKU not found' for invalid ones;<br>Fail: System returns incorrect coordinates or fails to find existing SKUs. | Critical for operators to quickly locate items for picking, reducing retrieval times.                    |
| FR-005  | Functional     | The system shall automatically trigger an alert notification to the Logistics Supervisor if a barcode/RFID scan detects an item placed in an incorrect rack position. | Medium | Pass: Alert is generated and displayed on the supervisor's dashboard within 5 seconds of the misplacement;<br>Fail: Misplacement occurs without triggering an alert.                  | Prevents inventory inaccuracies and ensures misplaced items are resolved immediately.                    |
| NFR-001 | Non-Functional | The warehouse item lookup service must locate any SKU pallet coordinate across 100,000 bins in under 100 ms.                                                          | High   | Pass: Benchmarking tests confirm target latency of < 100 ms under simulated peak load of 1,000 concurrent requests;<br>Fail: Average search lookup response time exceeds 100 ms.      | Ensures warehouse operations are not delayed by slow system responses.                                   |
| NFR-002 | Non-Functional | The warehouse inventory tracking system shall maintain 99.9% uptime during operational hours (6:00 AM to 10:00 PM).                                                   | High   | Pass: System monthly downtime during operational hours is less than 30 minutes;<br>Fail: System is unavailable for more than 30 minutes in a month.                                   | High availability is required to prevent halting warehouse loading/unloading operations.                 |

2. UML Use-Case Diagram

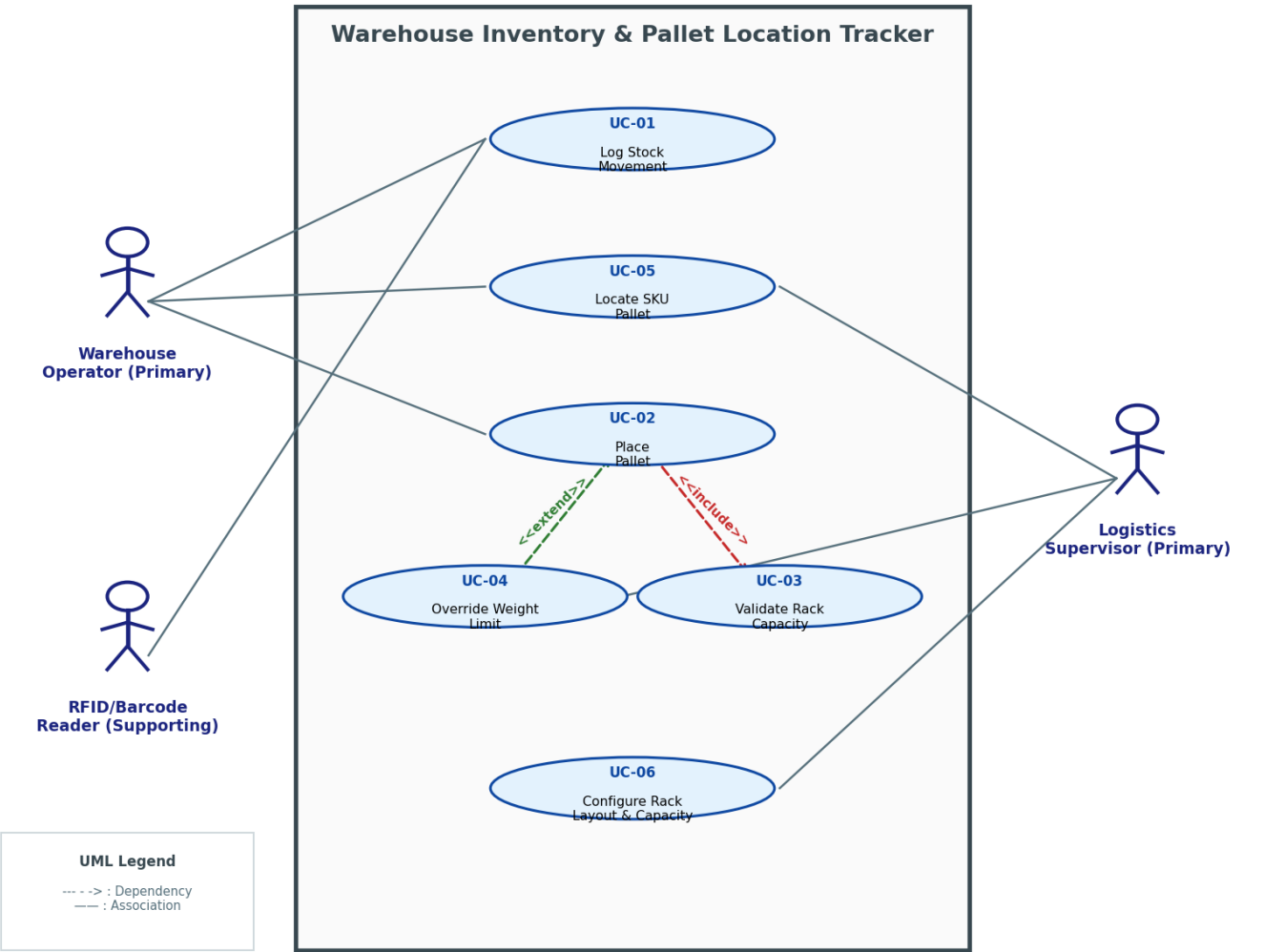


Figure 1: UML Use-Case Diagram for Warehouse Pallet Tracker (Problem Statement #27). The diagram models three actors: Warehouse Operator (Primary), Logistics Supervisor (Primary), and the RFID/Barcode Reader (Supporting hardware actor). Key relationships include the <<include>> relationship from 'Place Pallet' to 'Validate Capacity', and the <<extend>> relationship from 'Override Weight Limit' to 'Place Pallet'.

### 3. Use-Case Flow Document: Place Pallet (UC-02)

#### Preconditions:

1. The Warehouse Operator is logged into the handheld terminal.
2. The pallet is loaded onto a forklift/pallet jack.
3. The target rack and coordinates are defined and currently empty.

#### Postconditions:

1. The pallet is successfully placed at the 3D coordinates.
2. The system updates the rack's current aggregate weight.
3. The stock location is updated in the database.

#### Main Success Scenario (Step-by-Step):

1. Operator scans the barcode/RFID tag on the pallet.

2. Sy

#### Alternate Flow (8a. Overweight / Capacity Validation Failed):

- 4a. System detects that the proposed pallet weight exceeds the rack's maximum load capacity.
- 4b. System displays an 'Overweight Alert' and blocks the placement.
- 4c. Operator can either:
  - Choose a different rack slot (re-initiates step 3).
  - Request a Logistics Supervisor override (initiates UC-04: Override Weight Limit). If approved, placement proceeds to step 5.
  - Cancel the operation (operation terminates, pallet remains in original state).